Letter of Application

October 31, 2026

Members of the Search Committee
Department of Mathematics and Computer Science
Susquehanna University
Selinsgrove, Pennsylvania

Re: Assistant Professor of Computer Science, Job 0000311

Dear Members of the Search Committee:

I am applying for the tenure-track Assistant Professor of Computer Science position beginning in August 2027. I earned a PhD in Electrical and Computer Engineering from the University of Oklahoma, where I worked on machine learning for wireless networks. My path into computer science has been shaped by two things I enjoy equally: helping students learn to program and using computation to answer difficult questions about real systems. Susquehanna's emphasis on undergraduate teaching, faculty-guided research, and individual capstone projects is the kind of setting in which I would like to build my academic career.

At the University of Lahore, I taught Computer Programming, Communication Systems, and Wireless Communication, and advised student projects. Earlier, I supported communication systems laboratories and assessment at the University of Engineering and Technology, Taxila, and later mentored Bachelor's and Master's research students at COMSATS University Islamabad. I also have broader experience tutoring students outside the university classroom. Across these settings, I learned that the most useful feedback often begins with a simple question: where did the student's reasoning first diverge from the problem? In programming, I try to make that reasoning visible through small examples, code tracing, incremental debugging, and explanations of why one solution is more reliable or efficient than another.

My strongest areas for undergraduate CS teaching are introductory programming, computational problem-solving, data science, computer and communication networks, and applied machine learning. I would also work closely with colleagues to prepare for other core courses assigned within the department. Susquehanna's B.S. curriculum includes a progression from Principles of Computer Science and Object Oriented Programming and Design to Data Structures, Algorithms, and senior-level work. I would approach that progression by emphasizing conceptual connections, meaningful programming practice, and transparent assessment, rather than treating technical tools as a substitute for a strong foundation in computing.

My scholarship examines what happens when a machine-learning model encounters data unlike the examples on which it was trained. In my doctoral work, I studied data-efficient propagation modeling, wireless network resilience, and optimization under changing conditions. Related research uses smartwatch sensor data to study activity recognition and the limitations of models trained across small groups of users. At AT&T Labs, Futurewei, and AT&T, I have seen how these questions emerge in practice: imperfect labels, changing operating conditions, and the need to make a result useful to an engineer. I would bring these experiences into manageable student research projects that begin with a clear baseline and end with defensible conclusions.

I could envision individual Susquehanna capstones such as a Python tool that detects anomalies in public network traffic traces, a study comparing simple and deep learning models when training data are limited, or a wearable-activity classifier accompanied by a careful analysis of false positives and privacy considerations. Each could be scaled to the student's background and include a literature review, version-controlled implementation, evaluation against a baseline, and a final report or demonstration. These are projects that invite a student to make meaningful decisions rather than simply reproduce an existing model. They could also connect interested students to my longer-term research on reliable, responsible learning from real-world data.

A liberal arts college offers an opportunity to teach computation as both a technical discipline and a way of investigating problems across fields. I value students who ask questions from perspectives different from my own, and I would

welcome collaborative projects with colleagues and students working in the sciences and other disciplines. My aim would be to sustain a focused research program that contributes to the field while giving undergraduates genuine ownership of a part of the work.

Thank you for considering my application. I have included my curriculum vitae, teaching philosophy, research plan, and reference information. I would welcome the opportunity to discuss how my teaching experience, computing background, and research plans would contribute to Susquehanna's students and Department of Mathematics and Computer Science.

Sincerely,

Waseem Raza, PhD

waseem.placements@outlook.com | waseemraza844@gmail.com | +1 732-803-9703